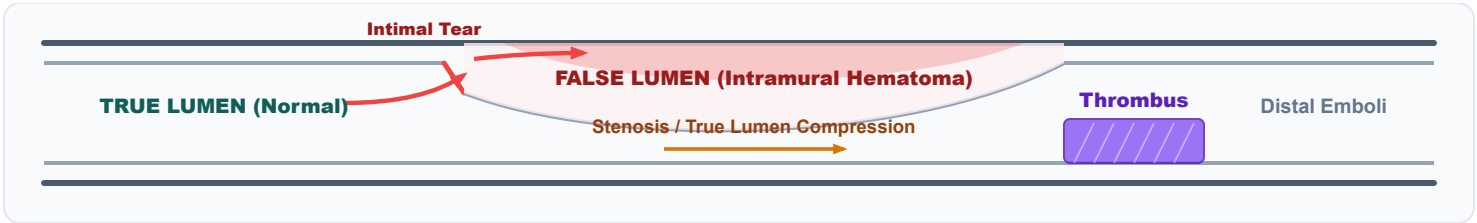


# Cervical Artery Dissection

Clinical presentation, diagnostic workup, medical management (extracranial vs. intracranial), and trial evidence.



## 1. Clinical Presentation & Symptoms

### Ipsilateral Pain (Local)

- **Carotid (ICA):** Frontotemporal, orbital, or facial headache/pain.
- **Vertebral (VA):** Occipital or posterior neck pain.
- **Onset:** Usually precedes ischemic stroke by hours to days (median 4d).

### Focal Neurological Signs

- **Horner Syndrome (~28–58%):** Ptosis & miosis. Anhidrosis is **absent (spared)** because sweat fibers travel on external carotid artery plexus.
- **Lower Cranial Nerve Palsies (~8–16%):** Local mass effect or ischemia of vasa nervorum (CN IX, X, XI, XII; tongue deviates to dissection side).

## 2. Diagnostic Workup

- **CTA/MRA Head & Neck:** Primary screen. Shows tapered stenosis ("string sign"), occlusion, flap, or pseudoaneurysm.
- **MRI Neck (T1 Fat-Sat):** Visualizes crescent-shaped hyperintense intramural hematoma (pathognomonic).
- **DSA:** Gold standard but invasive. Reserved for diagnostic doubt or endovascular stenting.

## 3. Medical Management: Extracranial vs. Intracranial Dissection

### Extracranial Dissection

- **Antithrombotics for ≥ 3 Months:** Strongly indicated (Class I, LOE C-LD) to prevent ischemic stroke/embolism.
- **Antiplatelet vs. Anticoagulation:** RCTs confirm clinical equipoise (no difference in stroke or death). Option between antiplatelet monotherapy/DAPT vs. anticoagulation (VKA or DOAC) is individualized.
- **Stenting:** Reserved for recurrent ischemia despite maximal medical therapy or expanding pseudoaneurysm.

### Intracranial Dissection (Special Caution)

- **High Subarachnoid Hemorrhage (SAH) Risk:** Intracranial arteries lack an external elastic lamina and have a thin adventitia; dissection can cause aneurysmal rupture & fatal SAH.
- **Anticoagulation Caution/Contraindication:** Anticoagulation is contraindicated if SAH is present.
- **Mandatory Screening:** Must rule out SAH with baseline CT Head or LP before starting antithrombotics.
- **Preferred Agent:** Antiplatelet monotherapy is generally preferred for intracranial dissection if no SAH is present.

## 4. Landmark Trial Evidence

| Trial                 | Population (N)                                                              | Interventions                                                                    | Primary Outcome & Stroke Rate at 14d/3mo                                                                                                                                                                                                                                                                                                                                   |
|-----------------------|-----------------------------------------------------------------------------|----------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>CADISS 2015</b>    | N = 250 patients with extracranial dissection (96% carotid, 4% vertebral).  | Antiplatelets (n=126) vs. Anticoagulants (n=124) within 7 days of onset.         | <b>Primary Endpoint (Stroke/Death at 3 Months):</b> <ul style="list-style-type: none"><li>• <b>Antiplatelet:</b> 2.0% (2/126)   <b>Anticoagulant:</b> 1.0% (1/124)</li><li>• <b>No difference</b> (p=0.63). Supported clinical equipoise.</li></ul>                                                                                                                        |
| <b>TREAT-CAD 2021</b> | N = 194 patients with extracranial dissection (per-protocol analyzed: 173). | Aspirin 300 mg daily (n=91) vs. VKA (INR 2.0–3.0, n=82) within 14 days of onset. | <b>Composite Endpoint (Stroke, Major Bleed, Death, or MRI Lesion at 14d):</b> <ul style="list-style-type: none"><li>• <b>Aspirin:</b> 23% (21/91)   <b>VKA:</b> 15% (12/82) [Diff 8%, 95% CI -4% to 21%]</li><li>• <b>Non-inferiority NOT met</b> (upper CI limit 21% &gt; 12% margin).</li><li>• <b>Clinical Stroke:</b> Aspirin 8.0% (7/91) vs. VKA 0% (0/82).</li></ul> |

**CADISS Trial:** CADISS Investigators. *Lancet Neurol.* 2015;14(4):361-7. PMID: 25684164 | **TREAT-CAD Trial:** Engelter ST, et al. *Lancet Neurol.* 2021;20(5):341-350. PMID: 33765420  
**AHA/ASA 2021 Guideline:** Kleindorfer DO, et al. *Stroke Prevention.* *Stroke.* 2021;52:e364-e467. PMID: 34024117